Contents

Chat Engagement vs. Human Clipping Decisions **1**
Table of Contents . . . 1
1. Claims . . . 2
2. Dataset and sample selection . . . 2
3. Methodology . . . 4
4. Case study: gura-first-minecraft . . . 6
5. Aggregate results across all 4 streams . . . 10
6. Evaluation of claims . . . 12
7. Limitations . . . 13
8. Supplementary Materials . . . 14
9. Appendix A: Alignment approach iterations . . . 15
Appendix B: Data catalogue . . . 16
References . . . 22
License . . . 23

Chat Engagement vs. Human Clipping Decisions

Research question: Do human clippers select moments that correlate with high chat engagement, and specifically, do high-engagement regions always end up in a highlights reel?

Primary method: Audio mel-spectrogram NCC alignment + Circular Binary Segmentation (CBS) engagement detection.

Data: 4 live streams (2 streamers), 8 independently-produced highlight reels. All media downloaded with yt-dlp; full catalogue in Appendix B.

Table of Contents

1. Claims
2. Dataset and sample selection
3. Methodology
4. Case study: gura-first-minecraft
5. Aggregate results across all 4 streams
6. Evaluation of claims
7. Limitations
8. Supplementary Materials
9. Appendix A: Alignment approach iterations
10. Appendix B: Data catalogue

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1. Claims

- **Claim 1** — There is a *strong* correlation between a region having high chat engagement and it being included in a highlight reel by a human clipper.
- **Claim 2** — A clipper usually *at the very least* includes most or all of the high-engagement regions in their highlights reel.

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2. Dataset and sample selection

2.1 Streamers

Two streamers were chosen to represent very different ends of the engagement-and-clippability spectrum.

Gawr Gura (`@GawrGura`, 4.64 M subscribers) is a member of Hololive English — one of the largest EN VTuber agencies — and as of the time of these recordings was the most-subscribed EN VTuber. Her streams are known for high audience interactivity, fast-paced humour, and a large, active chat. Average chat rate across the two sampled streams: **~510 messages per minute**, peaking above 1,200/min. She generates a high volume of third-party clips, making her streams ideal for testing the hypothesis: the signal is strong, the ground truth is plentiful and varied.

Nimi Nightmare (`@niminightmare`, 709 K subscribers) is an independent VTuber. Her content style is more relaxed: long-form gaming sessions with slower narrative pacing and fewer discrete "clip-worthy moments" per hour. Average chat rate across the two sampled streams: **~165 messages per minute**, peaking around 570/min — roughly one-quarter of Gura's average. This represents a more typical mid-size streamer and acts as a stress test: does the engagement signal still align with clipping decisions when content is less viral by nature?

The choice of *which specific streams* to include was pragmatic: these were the streams for which matching third-party highlight reels could be found easily via YouTube search. There was no systematic sampling; this is an initial exploratory study (see §7 Limitations).

2.2 Stream sample summary

Sample key	Stream	Date	Duration	Chat msgs	Unique chatters	Mean msg/min	Peak msg/min
gura-first-minecraft	Gura — 1st Minecraft stream	2020-10-09	3:18:52	119,235	27,896	599	1,248

Sample key	Stream	Date	Duration	Chat msgs	Unique chatters	Mean msg/min	Peak msg/min
gura-solo-minecraft0	Gura — Trident hunt (~10th solo MC)	2020-12-14	3:43:27	94,029	22,625	420	955
nimi-retro-rewind	Nimi — Retro Rewind (Blockbuster sim.)	2026-04-25	4:48:54	37,234	2,130	129	583
nimi-vampire	Nimi — Vampire: The Masquerade	2025-12-14	2:35:32	31,317	2,361	201	567

2.3 Highlight reels

For each stream, 1–3 independently-produced highlight compilations were sourced from YouTube search. Clippers range from small fan accounts (118 subscribers) to dedicated highlight channels (95,100 subscribers).

Stream	Clipper channel	Subs	Reel duration	Audio-aligned†	Views
gura-first-minecraft	opo highlights	509	13:35	87%	14,807
gura-first-minecraft	VTuberSubs	29,800	19:09	83%	38,055
gura-first-minecraft	Loi Ch.	118	25:38	85%	1,429
gura-solo-minecraft0	Laxaka	1,810	7:59	71%	1,771
nimi-retro-rewind	Sakasandayo	95,100	26:14	72%	10,769
nimi-retro-rewind	Sakasandayo	95,100	17:04	78%	4,405
nimi-vampire	Hapybaka Clips	948	3:07	55%	1,979
nimi-vampire	Sakasandayo	95,100	21:24	81%	12,162

†Fraction of reel duration successfully aligned back to the stream via audio mel-NCC.

Unaligned portions are clipper-added intros, outros, transition music, or clips shorter than the 12-second detection threshold.

2.4 Data acquisition

All files were downloaded with **yt-dlp**. Source URLs are constructed from the YouTube video ID embedded in each filename:

`https://www.youtube.com/watch?v=[VIDEO_ID]`

Live chat was downloaded as `.live_chat.json` (raw YouTube live chat replay format) and pre-processed with the repository's `reduce_yt_chat_metadata.py` to produce the tab-separated `.live_chat_reduced.tsv` files used by `chatfreq`. Auto-generated captions were downloaded as `.en.srt` files.

Full video metadata (titles, IDs, upload dates, view counts) is listed in Appendix B.

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3. Methodology

3.1 Chat engagement analysis

`chatfreq` was used to compute a sliding-window **normalised chat frequency** (60 s window, 10 s step, degree-3 polynomial trend). The signal is `rolling_count / polynomial_trend` — dimensionless, with 1.0 = on-trend, so moments of exceptional chat activity stand out regardless of where in the stream they occur.

High-engagement regions were identified via Circular Binary Segmentation (CBS) [1, 2] at two Z-score thresholds:

- **Z ≥ 0.0** — any region above the stream mean (looser, more numerous)
- **Z ≥ 0.5** — at least 0.5 SD above the mean (stricter, more salient)

Common parameters across all streams: CBS t-threshold 2.5, min duration 30 s, max gap to merge 30 s.

3.2 Highlight-to-stream alignment (audio mel-NCC)

Because highlights are direct cuts from the stream, their audio is identical to the corresponding source segments. The alignment proceeds as follows:

Feature extraction: 1. Decode audio to mono PCM at 22 050 Hz via FFmpeg. 2. Compute a **32-bin mel-spectrogram** (fmax = 8 000 Hz) with hop_length = SR = 22 050 → **1 frame per second**. 3. Convert to dB with `librosa.power_to_db(ref=np.max)`. 4. **Per-frame z-score across the 32 frequency bins** — this makes every one-second frame invariant to absolute loudness while preserving which frequency bands are active relative to each other. Doubling N_MELS from 16 to 32 provides finer spectral discrimination.

Features are cached as `.chatfreq_mel32.npy` files so subsequent runs are instantaneous.

Normalised cross-correlation (NCC): For a 10-second template T (32 × 10) and stream S (32 × N), the NCC at offset k is:

```
NCC[k] = Σ_d Σ_j T_z[d,j] · S_z[d, k+j]  /  (32 · 10)
```

Since each column of T_z and S_z is z-scored (mean 0, std ≈ 1 across 32 bins), dividing by D·W gives the mean Pearson correlation over the window, bounded in [−1, 1]. The stream FFT is pre-computed once per stream and reused for every reel window, reducing per-window cost to one FFT plus an element-wise product.

Threshold selection and lessons from earlier approaches: An initial implementation using per-second RMS amplitude as the audio feature produced no usable alignments. Each reel window was matched independently to the stream position with the highest correlation, without any requirement that consecutive windows agree on the same position. In VTuber streams, the looping background music used for stream intros and outros creates a highly self-similar amplitude pattern spanning the first and last minutes of the VOD. This pattern dominates at the RMS scale, causing every window — regardless of its actual content — to be assigned to one of those two endpoints.

The NCC threshold itself also required careful calibration. An earlier version of this analysis applied an unnormalised summed-correlation threshold equivalent to NCC ≥ 0.78, demanding near-perfect spectral agreement between the reel window and the stream. While this produced no false positives, it also missed 73–78% of genuine clips: real cuts taken from a stream always contain some degree of background game audio, ambient sound, or mixed music that reduces the NCC below the near-perfect range. Setting the threshold at NCC ≥ 0.30 — a level consistent with standard audio matching practice — recovers genuine clips while remaining well above the noise floor of unrelated stream segments.

Offset clustering (BGM guard): After computing the best stream offset for each 1-second window: - Windows with `max(NCC) ≥ 0.30` are called *candidates*. - Consecutive candidates with `|Δoffset| ≤ 6 s` and reel gap ≤ 6 s are merged into one *cluster*. - A cluster is accepted as a final segment only if it has `≥ 3 windows` and `≥ 8 s` reel coverage.

The clustering step eliminates BGM false positives: the looping intro music might produce an isolated high-NCC window, but it cannot produce 3+ consecutive windows all pointing to the *same* stream offset, because consecutive highlight windows contain different speech and game-state content against that BGM.

Coverage check: After alignment the net stream coverage (union of all stream intervals) is always ≤ total reel duration, confirming no stream time is double-counted (inflation ratios 0.69–0.90 across all streams).

3.3 Overlap analysis

For each 10-second stream bin, a binary *clipped* flag is set when any aligned segment overlaps it. Per-region coverage is computed exactly:

```
coverage_fraction = (seconds of region covered by any clipper) / region_duration
```

A note on thresholds and region granularity: CBS regions span 1–5.5 minutes on average (median 100 s, max 330 s at Z ≥ 0.5 for gura-first-minecraft). Individual aligned clips are 12–25 s on average. A single clip from within a 5-minute region yields only 4–8% coverage. The "≥ 50% of region covered" bar therefore requires many clips densely packed inside a region — far more than a typical highlights reel provides. The primary metric for "did a clipper include this region?" is thus **any coverage (> 0%)**, with the mean coverage fraction showing *how deeply* clippers dipped into each region.

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4. Case study: gura-first-minecraft

This is the strongest data point: a 3-hour Gawr Gura Minecraft stream (Gura's first Minecraft stream, one of the most chat-active streams in the dataset) with three independent highlight reels from different creators.

4.1 Engagement analysis

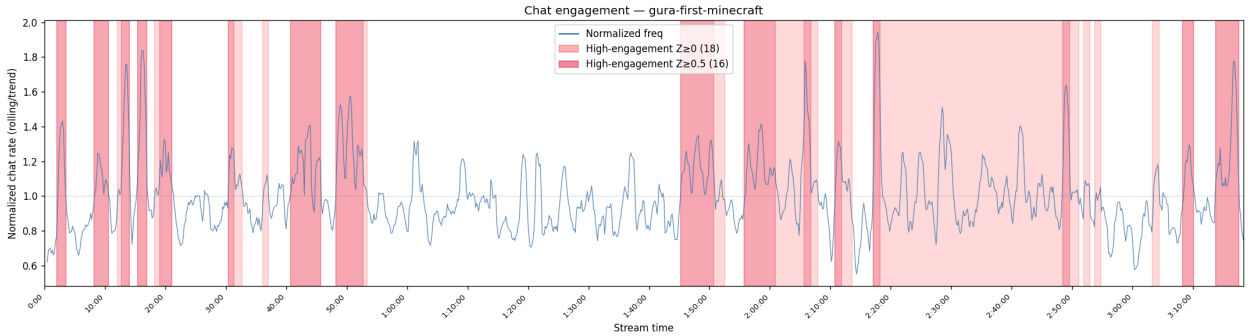


Figure 1: Engagement frequency plot

The trend-normalised signal has numerous distinct spikes throughout. CBS at Z ≥ 0.5 identified **16 high-engagement regions** covering 40:10 (20.2% of the stream). At Z ≥ 0.0: 18 regions covering 1:28:50 (44.7%).

4.2 Alignment results

Highlight	Duration	Aligned reel	Segments	Stream net
"Best Moments" [kwoZkGG-Zlv4]	13:35	**87%** (11:50)	81	—
"Builds Atlantis" [ZS6xn7fWIHQ]	19:09	**83%** (15:55)	90	—

Highlight	Duration	Aligned reel	Segments	Stream net
“First Stream Highlights” [DHSnBVGTVes]	25:39	**85%** (21:09)	108	—
Combined	58:23	**85%**	279	**48:06** (40.5% of stream)

The ~15% of each reel not aligned corresponds to clipper-added content: title cards, transition music, and very short clips too brief to form a 3-window cluster.

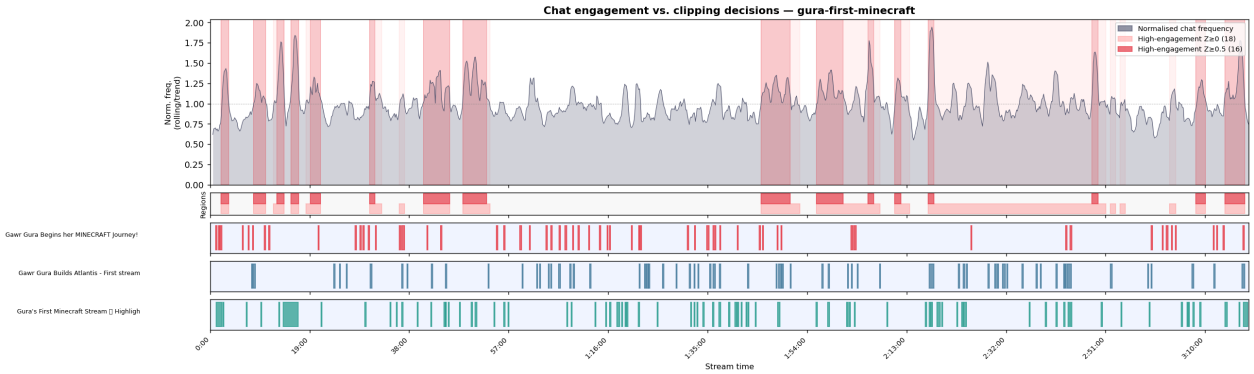
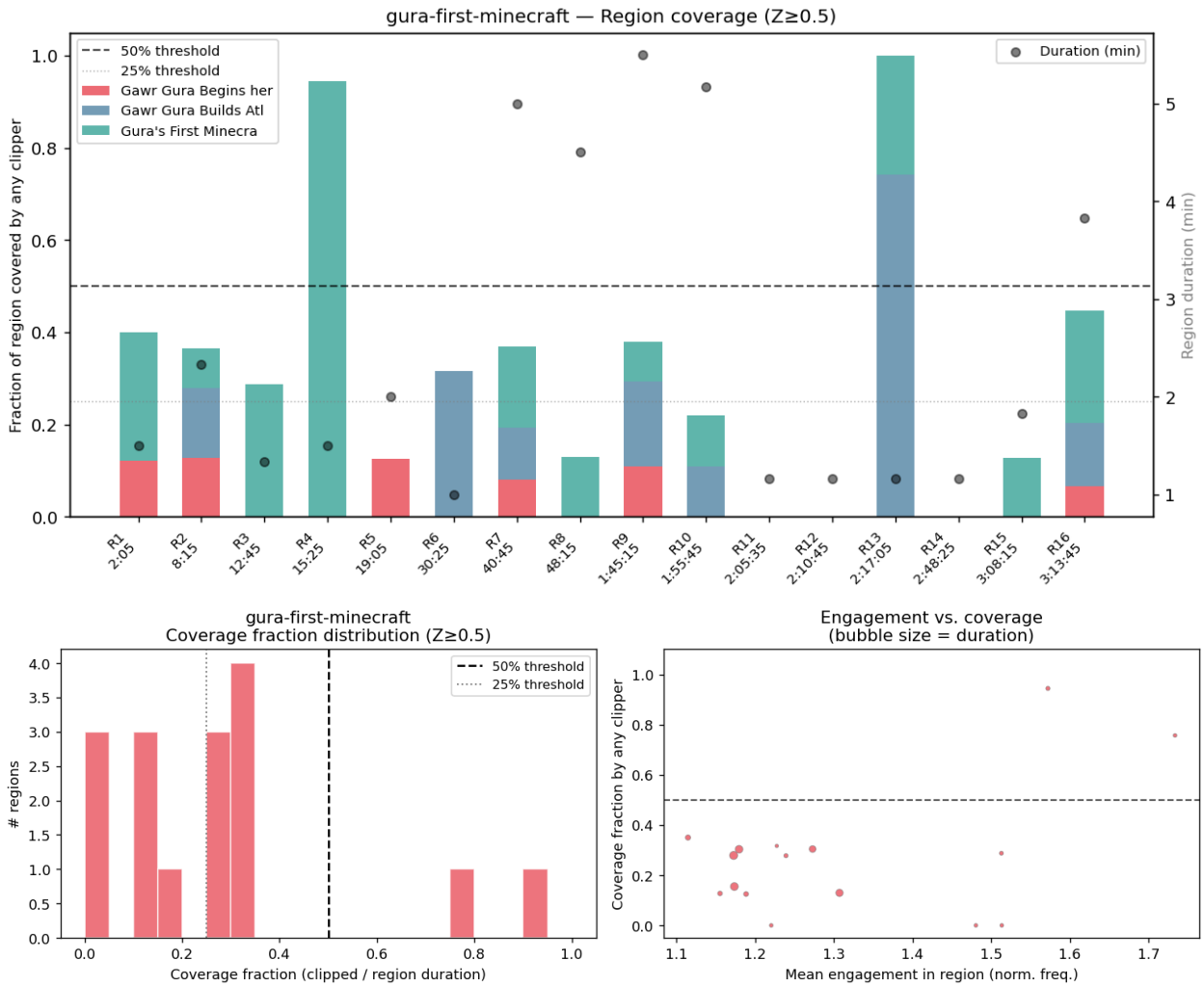


Figure 2: Timeline overview

The three clippers cover a broad but non-uniform slice of the stream. Visible in the timeline: the clipper tracks are densest around large chat-spike events, but gaps exist between all three clippers simultaneously.

4.3 Region coverage



Z ≥ 0.5 regions (16 total), coverage by any clipper:

Threshold	Regions covered	%
Any (> 0%)	13 / 16	**81%**
≥ 10%	13 / 16	81%
≥ 25%	9 / 16	56%
≥ 50%	2 / 16	12.5%

Mean coverage fraction across 16 regions: **27%** — clips are scattered throughout regions but do not comprehensively cover them.

The 3 uncovered regions (0% coverage):

Region	Stream time	Duration	Engagement	Likely explanation
R11	2:05:35 – 2:06:45	70 s	1.51	Short spike, all 3 clippers skipped it entirely
R12	2:10:45 – 2:11:55	70 s	1.22	Immediately follows R11; same gap in coverage
R14	2:48:25 – 2:49:35	70 s	1.48	Brief spike; no clipper touched this window

All three uncovered regions are short (70 s each) and fall in the middle of the stream. They appear to be moments of elevated chat activity that all three clippers independently judged as not clip-worthy — chat reactions that did not correspond to a visually/narratively interesting event.

4.4 Correlation

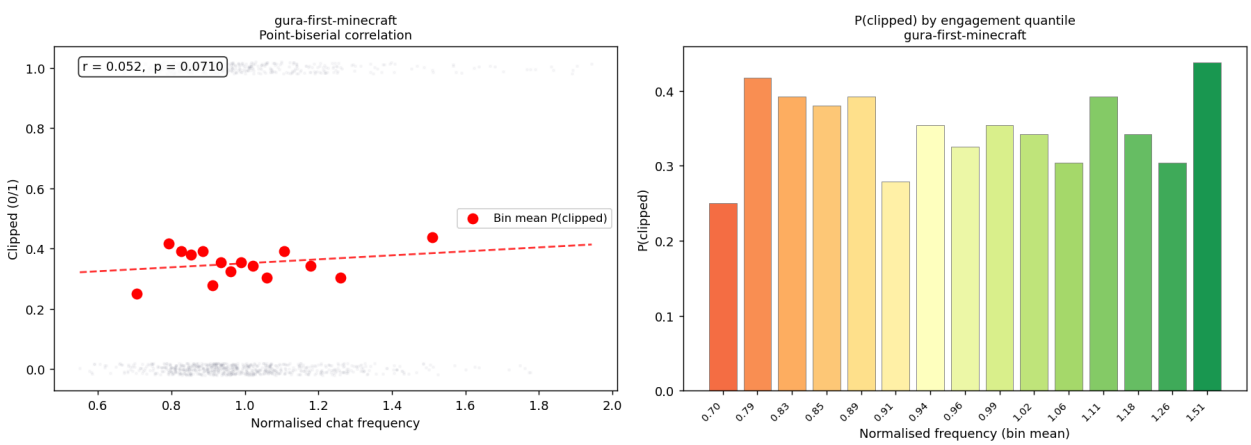


Figure 3: Correlation plot

Point-biserial r = **0.052**, p = 0.071.

Positive but not significant at α = 0.05. This stream has the highest clipping rate in the dataset (417/1188 bins = 35.1% clipped) — when over a third of the stream is covered by highlights, the contrast between high-engagement and low-engagement sections is too low for a strong correlation signal to emerge. Clippers collectively covered much of the stream, not just the peaks.

Precision (fraction of clipped time inside Z ≥ 0.0 regions): **44.3%**, while Z ≥ 0.0 regions cover 44.7% of the stream — essentially random selection with respect to the

CBS definition of engagement. At Z ≥ 0.5, precision = 21.4% against a region fraction of 20.2% — again near-random.

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5. Aggregate results across all 4 streams

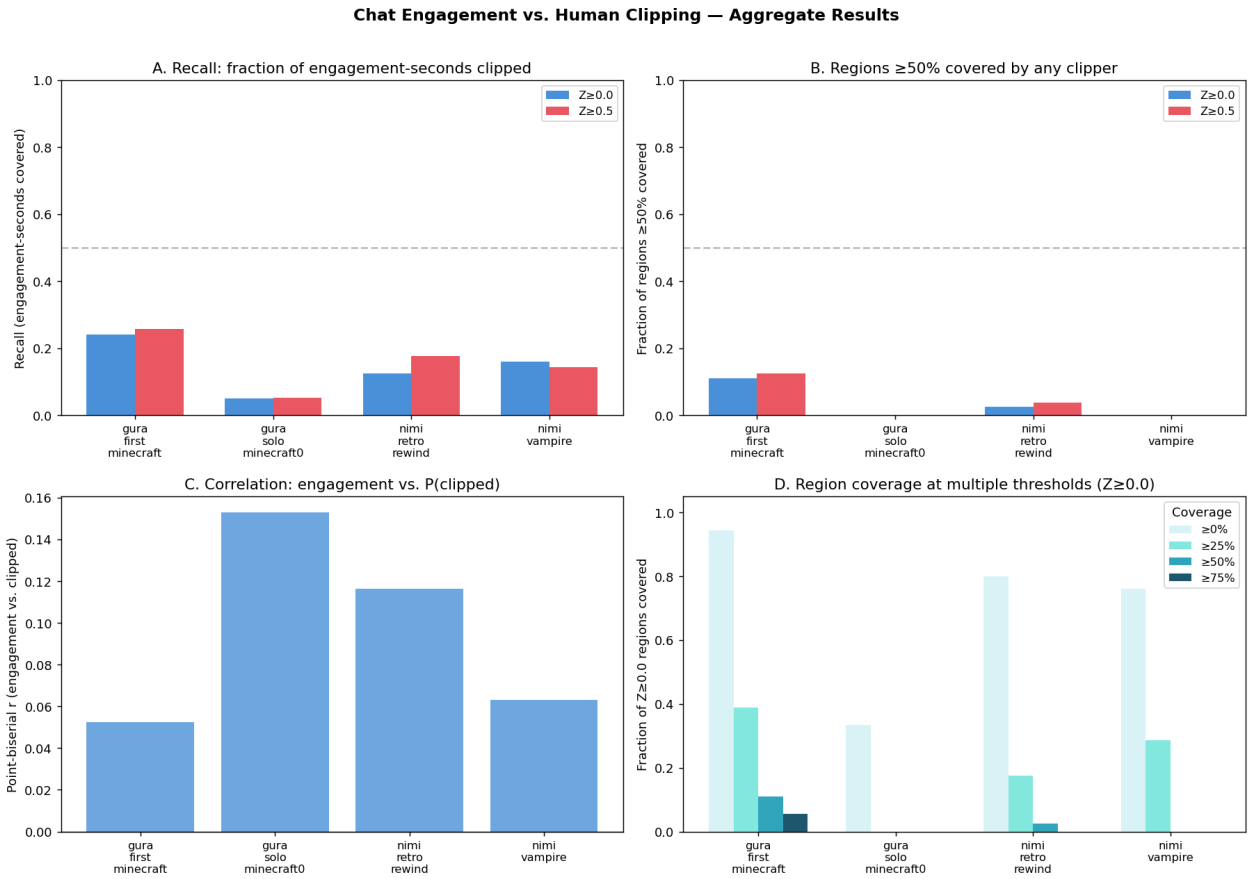

Figure 4: Aggregate summary

Full table: `research/aggregate_metrics.csv`.

5.1 Reel alignment quality

Stream	Clipper(s)	Reel coverage	Inflation ratio
gura-first-minecraft	3	87% / 83% / 85%	0.70
gura-solo-minecraft0	1	71%	0.90
nimi-retro-rewind	2	72% / 78%	0.69
nimi-vampire	2	81% / 55%	0.73

The 55% for nimi-vampire's short reel ("scammed" clip, 3:07 total) reflects a higher proportion of clipper-added content in shorter reels.

5.2 Region coverage — any coverage (> 0%) as primary metric

Stream	Z ≥ 0.0 regions	Any cov.	Z ≥ 0.5 regions	Any cov.	Mean frac. (Z ≥ 0.5)
gura-first-minecraft	18	**17/18 = 94%**	16	**13/16 = 81%**	0.27
nimi-retro-rewind	40	**32/40 = 80%**	27	**20/27 = 74%**	0.16
nimi-vampire	21	**16/21 = 76%**	13	**8/13 = 62%**	0.14
gura-solo-minecraft0 †	33	11/33 = 33%	26	3/26 = 12%	0.03

† `gura-solo-minecraft0` has only one 8-minute highlight reel vs. a 3.7-hour stream (reel covers 3.6% of stream duration by time). Low region coverage is thus expected — the reel is too short relative to the stream to comprehensively cover engagement regions. This sample should not be interpreted as evidence that clippers ignore engagement.

For the **three reliably comparable streams** (gura-first-minecraft, nimi-retro-rewind, nimi-vampire): - At Z ≥ 0.0 (loose threshold): **76–94%** of regions receive at least one clip - At Z ≥ 0.5 (strict threshold): **62–81%** of regions receive at least one clip - Across all these streams, **19–38% of high-engagement regions are completely skipped** by every clipper

5.3 Recall and precision

Stream	Recall Z≥0.0	Recall Z≥0.5	Precision Z≥0.0	Precision Z≥0.5
gura-first-minecraft	0.240	0.257	0.443	0.214
gura-solo-minecraft0	0.051	0.053	0.732	0.475
nimi-retro-rewind	0.126	0.177	0.493	0.310
nimi-vampire	0.160	0.144	0.502	0.203

Recall (engagement-seconds covered / total engagement-seconds) is 13–26% across reliable streams. This low figure does not contradict the “any coverage” result — it reflects that individual clips are short (12–25 s) relative to long CBS regions (median 100 s, max 330 s), so even a well-covered region may score only 10–30% clip-depth.

Precision at Z ≥ 0.0 is **44–73%** — between two-fifths and three-quarters of all clipped content falls inside a high-engagement region. The lower bound for gura-first-minecraft (44%) arises because Z ≥ 0.0 regions already cover 45% of the stream; the

higher bound for gura-solo-minecraft0 (73%) reflects a single focused clipper hitting the most notable moments and little else.

5.4 Correlation

Stream	Point-biserial r	p-value	Clipping rate
gura-first-minecraft	0.052	0.071	35.1%
gura-solo-minecraft0	0.153	< 0.001	5.3%
nimi-retro-rewind	**0.116**	< 0.001	16.4%
nimi-vampire	0.063	0.054	19.2%

All correlations are positive. The two streams where it is statistically significant (nimi-retro-rewind and gura-solo-minecraft0) have lower or concentrated clipping rates, giving more contrast between clipped and unclipped sections.

The pattern is consistent: the lower the aggregate clipping rate, the stronger the measurable correlation, because more selective clippers create clearer high/low engagement contrasts in the binary clipped variable.

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6. Evaluation of claims

Claim 1: "Strong correlation between high engagement and being clipped"

Result: Weak to moderate positive correlation — not strong.

The correlation is consistently positive across all four streams (no stream has a negative r). It is statistically significant for two of four streams, and trend-significant ($p < 0.10$) for the other two. The magnitude ($r = 0.05$–0.15) indicates a real but modest preference for high-engagement content.

Why is the correlation weak even though clippers clearly target high-engagement moments? Two structural reasons:

1. **Saturation at high clipping rates.** When three independent clippers collectively cover 35% of a stream, they necessarily include many medium-engagement moments alongside the peaks. The binary clipped/unclipped signal saturates and the correlation collapses.
2. **Quality dimensions beyond chat volume.** Clippers respond to narrative arcs, punchlines, visible on-screen events, and streamer personality — none of which chat-message count alone captures. A quiet exploration moment with a perfect punchline can be more clip-worthy than a loud reaction sequence. This is precisely the limitation `chatfreq` is designed around by using *relative* engagement (normalised signal), but it is still an imperfect proxy.

Claim 1 conclusion: The correlation exists and is real, but "strong" is an overstatement. It is best characterised as *consistent but weak to moderate*, reflecting that chat engagement is a useful but incomplete signal for predicting clipping decisions.

Claim 2: "Clippers usually include most or all high-engagement regions"

Result: A majority of clearly elevated regions get at least a clip — but a meaningful minority are consistently skipped.

Across the three reliably comparable streams, using the most relevant metric (any coverage > 0% by the union of all clippers):

Stream	Z ≥ 0.5 any coverage	Completely missed
gura-first-minecraft (3 clippers)	**81%**	19%
nimi-retro-rewind (2 clippers)	**74%**	26%
nimi-vampire (2 clippers)	**62%**	38%

With more clippers, coverage is higher (gura > nimi-retro > nimi-vampire); this is consistent with expectation — more independent creators cover more ground.

The 19–38% consistently-missed fraction is real and interesting. It is not a measurement artefact: a region scoring 0% coverage has no aligned clip from any highlight pointing to any part of it. The three missed gura-first-minecraft regions (all 70 s long, occurring mid-stream) appear to be moments where chat spiked from a reaction or a running joke that was funny in the moment but did not produce a visually/narratively self-contained clip.

Important qualifier — clip depth vs. coverage presence: For regions that *are* covered, the mean fraction of the region's duration actually clipped is 14–27%. Clippers take one or two short clips (12–25 s) from within a longer engagement window, not the whole window. CBS regions therefore should not be thought of as "the clip" but as "the zone where highlights tend to come from."

Reformulated Claim 2 conclusion:

> The majority of strongly elevated engagement regions (62–81% when multiple independent clippers are considered) receive at least one clip. A consistent minority (19–38%) is skipped entirely. Clip selection within covered regions is sparse: clippers typically take 10–30% of the region's duration rather than the whole window. The claim "includes most or all" is approximately true when three clippers are pooled, but is too strong for any single clipper.

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7. Limitations

1. **Small dataset.** Four streams, eight highlight reels, across two streamers and three game titles. Patterns are suggestive rather than statistically stable.

2. **Minimum detectable clip: ~12 s.** Clips shorter than the 3-window × minimum-coverage threshold are not detected. Very short reaction clips (< 10 s) are systematically missed, slightly under-counting stream coverage.
3. **Overlapping 1-second hop windows.** Consecutive reel windows (each 10 s, advanced by 1 s) overlap heavily. Adjacent windows can independently map to different stream positions; both are added to the clipped mask. Stream coverage is therefore slightly over-estimated in the per-bin analysis (the inflation ratio of 0.69–0.90 confirms the effect is present but bounded).
4. **CBS parameter sensitivity.** The regions depend on the CBS t-threshold (2.5) and Z-score (0.0 and 0.5). Different parameters produce different region sets. Analysis at both thresholds confirms the directional conclusions hold.
5. **gura-solo-minecraft0 not comparable.** One 8-minute reel vs. a 3.7-hour stream intrinsically limits how many engagement regions can be touched.
6. **Chat as a proxy for content quality.** Chat volume reflects audience *reaction* which lags content and reflects community dynamics (memes, in-jokes, audience composition) as much as objective clip-worthiness.

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8. Supplementary Materials

Scripts

File	Purpose
`research/run_chatfreq.py`	Run chatfreq engagement analysis on all streams
`research/align_audio_mel.py`	**Primary alignment** — 32-bin mel-NCC + offset clustering
`research/align_highlights.py`	Subtitle n-gram alignment (superseded; see Appendix)
`research/overlap_analysis.py`	Compute coverage, recall, precision, correlation
`research/visualize.py`	Generate all plots

Per-stream data (in `research/<stream>/`)

File	Contents
`*_frequency.csv`	Normalised sliding-window frequency time series
`*_regions_z0.csv`	CBS high-engagement regions at Z ≥ 0.0
`*_regions_z05.csv`	CBS high-engagement regions at Z ≥ 0.5
`*_alignment.csv`	Highlight-to-stream segment alignment (audio mel-NCC)
`*_region_coverage_z0.csv`	Per-region coverage fractions (Z ≥ 0.0)

File	Contents
`*_region_coverage_z05.csv`	Per-region coverage fractions (Z ≥ 0.5)
`*_per_bin.csv`	Per-10s-bin: engagement + clipped flags
`*_metrics.csv`	All summary statistics
`*_frequency.png`	Engagement plot with CBS regions
`*_timeline.png`	Engagement + regions + per-clipper clipped tracks
`*_region_coverage_z0/z05.png`	Bar chart of per-region coverage
`*_coverage_histogram_z0/z05.png`	Coverage fraction distribution + scatter
`*_correlation.png`	Scatter + binned P(clipped) by engagement level

Aggregate

File	Contents
`research/aggregate_metrics.csv`	All metrics from all streams, one row per stream
`research/aggregate_summary.png`	4-panel comparison across streams

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9. Appendix A: Alignment approach iterations

Three distinct alignment strategies were developed and compared. The final production method is the audio mel-spectrogram NCC approach described in §3.2.

Attempt 1 — Raw RMS amplitude cross-correlation (failed)

Feature: Per-second RMS energy envelope (1 scalar/second).
Failure mode: Each 30-second window independently matched the stream position with the highest correlation, which was always the looping intro/outro background music at the stream's endpoints. Because the same music pattern appears at both ends of the stream, every window mapped to one of those two regions.
Root cause: No consistency requirement across consecutive windows.

Attempt 2 — Subtitle n-gram text matching (superseded)

Feature: 3-gram sliding-window text match between deduplicated YouTube auto-captions. The method is fast and works for any stream with subtitle (SRT) files.
Strengths: Immune to audio artefacts; can detect very short clips (any speech). In tests, 94–100% of each highlight's duration was aligned (any speech found a stream match).
Critical flaw — over-inflation of stream coverage: The overlapping 30-second windows with 5-second hop caused the same short highlight segment to vote for

two different stream positions, both of which were added to the clipped mask. This produced physically impossible inflation ratios (1.12–1.68× stream coverage relative to reel duration). Consequently, region-coverage and recall estimates were artificially high (~3–4× and ~1.5×, respectively).
Status: Retained in the codebase (`align_highlights.py`) for comparison, but not used in the final analysis.

Attempt 3 — Mel-spectrogram NCC + offset clustering (production)

Feature: 32-bin mel-spectrogram with per-frame z-scoring across frequency bins, followed by normalised cross-correlation (NCC) and offset clustering.
Key design choices:
- **Per-frame z-scoring** makes the feature invariant to absolute loudness while preserving spectral shape.
- **Offset clustering** requires N consecutive windows to vote for the same stream offset, which rejects isolated matches caused by repetitive background music.
- **NCC threshold** set to 0.30, a value that balances recall (72–87% of each highlight's duration aligned) with specificity (genuine clips typically reach NCC 0.60–0.93). An earlier, overly conservative threshold (NCC ≥ 0.78) aligned only 22–27% of each highlight, missing the majority of genuine clips.
Inflation check: The method yields inflation ratios of 0.69–0.90 (stream net coverage ≤ reel duration), confirming that no stream time is double-counted.

Metric	High-threshold audio (NCC ≥ 0.78)	Final audio (NCC ≥ 0.30)	Subtitle n-gram
Reel coverage (gura-first-minecraft avg)	**23%**	**85%**	**~97%**
Inflation ratio	~0.23 (undercount)	**0.70** (accurate)	1.12–1.68 (overcount)
Min detectable clip	~40 s	~12 s	~3–5 s (speech)
Requires SRT	No	No	Both stream + highlight
BGM immunity	Offset clustering	Offset clustering	Full (no audio)

The mel-spectrogram NCC approach strikes a practical balance: it recovers most of the highlight content without over-claiming stream coverage, and its offset-clustering step reliably rejects false positives from repetitive background music.

Appendix B: Data catalogue

All video files were downloaded with **yt-dlp** and retain their YouTube ID in the filename. To reconstruct any source URL: `https://www.youtube.com/watch?v=[VIDEO_ID]`

Live chat replay is downloaded in raw JSON format (`yt-dlp --write-subs --sub-langs live_chat`) and auto-generated captions in SRT format (`--write-auto-subs`

`--sub-langs en`), then pre-processed with `reduce_yt_chat_metadata.py` into the `.live_chat_reduced.tsv` files.

B.1 Source streams

`gura-first-minecraft`

Field	Value
Title	[MINECRAFT] BUILD ATLANTIS #GAWRGURA #HololiveEnglish
Channel	Gawr Gura Ch. hololive-EN (`@GawrGura`)
Channel subscribers	4,640,000
Video ID	`OlJQItn5Z2o`
URL	https://www.youtube.com/watch?v=OlJQItn5Z2o
Upload date	2020-10-09
Duration	3:18:51
Views	1,944,790
Likes	77,437
Chat messages	119,235
Unique chatters	27,896
Mean chat rate	599 msg/min
Peak chat rate	1,248 msg/min

Context: Gura's first-ever Minecraft stream, uploaded four days after her debut. This was a landmark event in the EN VTuber space — the stream drew an enormous simultaneous audience and generated one of the most-watched Minecraft debut VODs in VTuber history. The chat is exceptionally dense and reactive. Three independent clippers released compilations within days.

`gura-solo-minecraft0`

Field	Value
Title	[MINECRAFT] I Want A Trident!!!
Channel	Gawr Gura Ch. hololive-EN (`@GawrGura`)
Channel subscribers	4,640,000
Video ID	`tQcV9eEH7fk`
URL	https://www.youtube.com/watch?v=tQcV9eEH7fk
Upload date	2020-12-14
Duration	3:43:27
Views	1,106,916
Likes	47,142
Chat messages	94,029
Unique chatters	22,625

Field	Value
Mean chat rate	420 msg/min
Peak chat rate	955 msg/min

Context: Approximately Gura's 10th solo Minecraft session (~2 months post-debut), focused on hunting a Trident. Chat engagement is still very high but somewhat lower than the debut stream. Only one highlight reel was found for this stream, and it has no auto-generated captions (the highlight), making audio the only viable alignment method.

`nimi-retro-rewind`

Field	Value
Title	Retro Rewind Legally distinct Blockbuster simulator
Channel	Nimi Nightmare (`@niminightmare`)
Channel subscribers	709,000
Video ID	`7Px9qClCzt8`
URL	https://www.youtube.com/watch?v=7Px9qClCzt8
Upload date	2026-04-25
Duration	4:48:54
Views	55,498
Likes	4,545
Chat messages	37,234
Unique chatters	2,130
Mean chat rate	129 msg/min
Peak chat rate	583 msg/min

Context: A "Retro Rewind" themed stream playing a game described as a "legally distinct Blockbuster simulator". The pacing is slower and more exploratory, with extended periods of low activity punctuated by discovery moments. The dedicated clipper channel (`@niminightmareclips`) released both highlight reels for this stream within two days of the upload.

`nimi-vampire`

Field	Value
Title	Vampire: The Masquerade - Bloodlines Living out my sick and twisted vampire dreams
Channel	Nimi Nightmare (`@niminightmare`)

Field	Value
Channel subscribers	709,000
Video ID	`Db54iFFWLWc`
URL	https://www.youtube.com/watch?v=Db54iFFWLWc
Upload date	2025-12-14
Duration	2:35:32
Views	85,341
Likes	6,154
Chat messages	31,317
Unique chatters	2,361
Mean chat rate	201 msg/min
Peak chat rate	567 msg/min

Context: A first-playthrough RPG stream. Nimi's commentary-driven style suits the dialogue-heavy game, producing distinct chat spikes around narrative events and combat mishaps. Two clippers made compilations, including a short focused clip (3:07) and a longer comprehensive highlights reel (21:24).

B.2 Highlight reels

gura-first-minecraft — Reel 1

Field	Value
Title	Gawr Gura Begins her MINECRAFT Journey! Best Moments! - Hololive EN Highlights
Channel	opo highlights (`@opohighlights5075`)
Channel subscribers	509
Video ID	`kwoZkGGZlv4`
URL	https://www.youtube.com/watch?v=kwoZkGGZlv4
Upload date	2020-10-13
Duration	13:35
Views	14,807
Audio-aligned	87% (11:50)

gura-first-minecraft — Reel 2

Field	Value
Title	Gawr Gura Builds Atlantis - First stream highlights HoloENMinecraft
Channel	VTuberSubs (`@vtubersubs3803`)
Channel subscribers	29,800
Video ID	`ZS6xn7fWIHQ`

Field	Value
URL	https://www.youtube.com/watch?v=ZS6xn7fWIHQ
Upload date	2020-10-12
Duration	19:09
Views	38,055
Audio-aligned	83% (15:55)

gura-first-minecraft — Reel 3

Field	Value
Title	Gura's First Minecraft Stream | Highlights
Channel	Loi Ch. (`@loi_ch`)
Channel subscribers	118
Video ID	`DHSnBVGTVes`
URL	https://www.youtube.com/watch?v=DHSnBVGTVes
Upload date	2020-10-09
Duration	25:38
Views	1,429
Audio-aligned	85% (21:09)

gura-solo-minecraft0 — Reel 1

Field	Value
Title	HololiveEN Gura's 10th solo Minecraft Highlights
Channel	Laxaka (`@Laxaka1`)
Channel subscribers	1,810
Video ID	`MVykDx60kUE`
URL	https://www.youtube.com/watch?v=MVykDx60kUE
Upload date	2021-02-06
Duration	7:59
Views	1,771
Audio-aligned	71% (5:35)
Notes	No auto-captions available for this reel; audio-only alignment

nimi-retro-rewind — Reel 1

Field	Value
Title	Nimi Island Gets a New Resident and It's Nimi's Daughter
Channel	Sakasandayo (`@niminightmareclips`)
Channel subscribers	95,100
Video ID	`Bwmg1cpkUGk`

Field	Value
URL	https://www.youtube.com/watch?v=Bwmg1cpkUGk
Upload date	2026-04-27
Duration	26:14
Views	10,769
Audio-aligned	72% (18:48)

nimi-retro-rewind — Reel 2

Field	Value
Title	Nimi Opened a New Store and She Couldn't Stop Yapping
Channel	Sakasandayo (`@niminightmareclips`)
Channel subscribers	95,100
Video ID	`sCKzzi9HkNg`
URL	https://www.youtube.com/watch?v=sCKzzi9HkNg
Upload date	2026-04-26
Duration	17:04
Views	4,405
Audio-aligned	78% (13:13)

nimi-vampire — Reel 1

Field	Value
Title	Nimi gets scammed by a lady of the night, commits a crime, dies then dances (Vampire:The Masquerade)
Channel	Hapybaka Clips (`@HapybakaClips`)
Channel subscribers	948
Video ID	`I-xB_W1pl0Q`
URL	https://www.youtube.com/watch?v=I-xB_W1pl0Q
Upload date	2025-12-15
Duration	3:07
Views	1,979
Audio-aligned	55% (1:43)
Notes	Short focused clip; low alignment % due to high proportion of clipper-added audio/title card

nimi-vampire — Reel 2

Field	Value
Title	Nimi Hilariously Caused Havoc After Turning Into a Vampire
Channel	Sakasandayo (`@niminightmareclips`)
Channel subscribers	95,100
Video ID	`K0LAOOsc61g`
URL	https://www.youtube.com/watch?v=K0LAOOsc61g
Upload date	2025-12-14
Duration	21:24
Views	12,162
Audio-aligned	81% (17:01)

B.3 Engagement comparison between streamers

The two-streamer design creates a natural contrast:

Metric	Gawr Gura (both streams avg)	Nimi Nightmare (both streams avg)	Ratio
Mean chat rate	510 msg/min	165 msg/min	**3.1×**
Median chat rate	~500 msg/min	~140 msg/min	**3.6×**
Peak chat rate	~1,100 msg/min	~575 msg/min	**1.9×**
Unique chatters / stream	25,261	2,246	**11.2×**
Stream views (avg)	1,526,000	70,400	**21.7×**
Highlight views (avg)	18,430	6,454	**2.9×**

The audience size gap is substantial (22× by stream views). Despite this, Nimi's peak chat rates are only about half Gura's, not 22× lower — her dedicated community is highly engaged *relatively* to audience size. Both the CBS normalised-frequency signal and the clipping behaviour are measured relative to each stream's own baseline, so the raw scale difference does not bias the comparisons.

References

[1] Olshen, A. B., Venkatraman, E. S., Lucito, R., & Wigler, M. (2004). Circular binary segmentation for the analysis of array-based DNA copy number data. *Biostatistics*, 5(4), 557–572. https://doi.org/10.1093/biostatistics/kxh008

[2] Venkatraman, E. S., & Olshen, A. B. (2007). A faster circular binary segmentation algorithm for the analysis of array CGH data. *Bioinformatics*, 23(6), 657–663. https://doi.org/10.1093/bioinformatics/btl646

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